Toward Federated Customized Neural Architecture Search for Remote Sensing Scene Classification

Jianzhao Li, *Member, IEEE*, Shanfeng Wang, *Member, IEEE*, Rui Yang, Maoguo Gong, *Fellow, IEEE*, Zhuping Hu, Ning Zhang, Kai Sheng, *Member, IEEE*, and Yu Zhou

***Abstract*— Remote sensing (RS) scenarios usually involve sensitive geographic information on national security and regional development. In the commonly used centralized machine-learning paradigm, data dispersed in various locations are concentrated and processed on a single server, which is prone to privacy leakage and data security concerns. Besides, it is difficult to solve the high heterogeneity of RS images by simply applying federated learning (FL) algorithms to scene classification. In this article, we formulate a federated remote sensing scene classification (FedSC) framework, and design a customized neural architecture search (CNAS) to achieve both global generality for multiparty collaborative distributed training and local specificity for personalized RS scene customization. The proposed FedSC is generalizable to be implemented in any manually designed networks, network pruning strategies, or NAS methods related to remote sensing scene classification (RSSC). While the designed CNAS not only achieves collaborative distributed training in protecting participant data privacy to obtain a generalized global model, but also provides a customized local model for each participant that is more in line with the characteristics of private RS scenarios. Overall, the proposed FedSC$_{\text{CNAS}}$ provides a novel federated collaborative training paradigm for RSSC in terms of data privacy, data heterogeneity, and personalized customization. Extensive analytical and comparative experiments on three benchmark RSSC datasets validate the versatility and effectiveness of our methods, and the proposed FedSC$_{\text{CNAS}}$ exhibits superior competitiveness compared to state-of-the-art methods.**

***Index Terms*— Federated learning (FL), neural architecture search (NAS), personalized collaboration, remote sensing (RS) scene classification.**

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Jianzhao Li and Kai Sheng are with Guangzhou Institute of Technology, Xidian University, Guangzhou 510555, China, and also with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an 710071, China (e-mail: lijianzhao@xidian.edu.cn; kaisheng@xidian.edu.cn).

Shanfeng Wang and Rui Yang are with the School of Cyber Engineering, Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an 710071, China (e-mail: sfwang@xidian.edu.cn; yangrui@stu.xidian.edu.cn).

Maoguo Gong is with the School of Electronic Engineering, Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an 710071, China, and also with the Academy of Artificial Intelligence, College of Mathematics Science, Inner Mongolia Normal University, Hohhot 010028, China (e-mail: gong@ieee.org).

Zhuping Hu and Yu Zhou are with the School of Electronic Engineering, Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an 710071, China (e-mail: zhupinghu@stu.xidian.edu.cn; zhouyu@xidian.edu.cn).

Ning Zhang is with Shanghai Aerospace Electronic Technology Institute, Shanghai 230051, China (e-mail: dzs.zhangning@gmail.com).

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I. Introduction

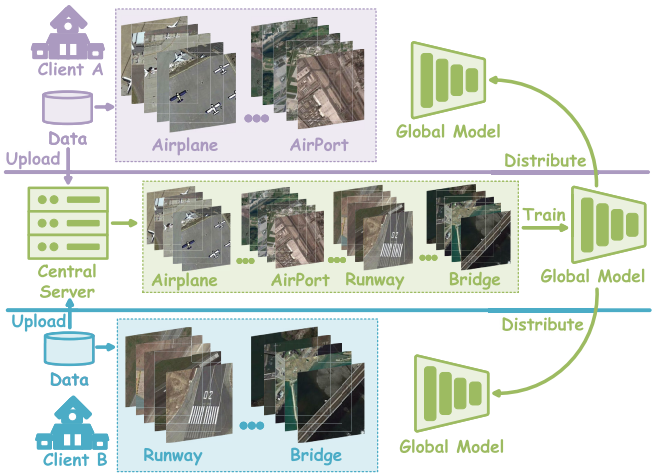
SCENE classification, an important hotspot in the field of remote sensing (RS), aims to automatically identify and classify surface scenes by analyzing and processing image data acquired from RS equipment (e.g., satellites, aircraft, and drones) [1], [2], [3], to provide decision-making support for geographic information systems [4], environmental monitoring [5], urban planning [6], and agricultural management [7].

With the acquisition of large-scale labeled RS scene data and the development of deep learning-based related techniques, scene classification has experienced unprecedented development [8]. In particular, neural network architectures such as convolutional neural networks (CNNs) [9], generative adversarial networks (GANs) [10], graph convolutional networks (GCNs) [11], and Transformer [12], training paradigms such as semi-supervised learning [13], self-supervised learning [14] and reinforcement learning [15], and improvement strategies such as data augmentation [16], attention modules [5] and feature fusion [17] have been widely studied. Nevertheless, most of the current remote sensing scene classification (RSSC) efforts are based on the centralized machine-learning training paradigm [18], i.e., RS data needs to be centralized on a server for training and inference as shown in Fig. 1(a).

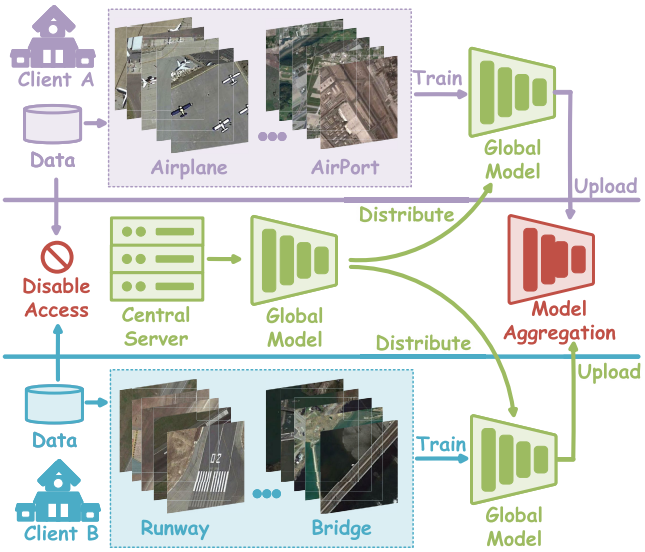
Although multiple parties involved in collaborative training can obtain a deep model with better generalization performance, the respective RS scene data are relatively sensitive and may contain user privacy, infrastructure or military facilities, and national security [19]. With the release of data regulatory policies in various countries and regions, data security has become a growing concern [20], [21]. In addition, due to the data privacy and security priority, many RS organizations are not willing to share their data, which leads to the problem of data silos, i.e., data are scattered across different systems and cannot be effectively integrated to be utilized [18], [22].

Federated learning (FL), an emerging distributed machine-learning framework as shown in Fig. 1(b), which permits multiple organizations to engage in collaborative learning without violating privacy protection, data security,

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(a)



(b)

Fig. 1. Comparison of centralized machine learning and FL paradigm. (a) Centralized machine-learning paradigm, all the clients need to centralize their private data on a central server, and the trained model is distributed back to the clients. (b) FL paradigm, all the clients do not have access to each other's data, but instead train the data locally to upload the model to the server for model aggregation.

and government regulations [23]. Recently, numerous outstanding researches have combined FL on different RS tasks, such as target recognition [24], radar detection [25], and RS image classification [22]. Although these works have contributed to the exploration of FL in the field of RS, most of the related work simply implements the task application and does not further explore the characteristics of RS data. For the RSSC, in real-world applications, the data held by different photogrammetry organizations are heterogeneous, i.e., different scenarios exhibit significant geographic heterogeneity in various geographic locations. Moreover, the local computing resources of each organization are also heterogeneous, and it is significantly inappropriate for real-world applications if the same network architecture is applied as in previous studies [20], [22], [24], [25], [26].

Therefore, there is an urgent need for a personalized customization strategy for RSSC in FL, which can provide each participant with a personalized architecture adapted to the local data. Overall, model customization for different data can be divided into network pruning strategies [27], [28], [29], [30], [31] and neural architecture search (NAS) [32], [33], [34], [35]. Network pruning strategies, similar to manually designed architectures, need to rely on certain expert knowledge to take targeted structured or unstructured pruning of different architectures, and involve multiple iterations and pruning operations, with relatively complex algorithmic implementations [36]. In contrast, NAS can automatically explore different architectures to obtain the optimal model. According to the search strategy specifically, NAS can be categorized as evolution-based, reinforcement learning-based, and gradient-based [37]. Evolution-based NAS is capable of obtaining the global optimal solution in both discrete and continuous search spaces, while reinforcement learning-based NAS enables sampling of effective architectures from the search space, but they both require significant computational resources and complex iterations [34], [37], [38], [39]. Notably, expensive communication and computational resources are not applicable in realistic federated RS applications. Gradient-based NAS, which utilizes the gradient descent method to differentiate the search space for efficient optimization, has been widely studied in the field of RS [36], [40], [41], [42]. It is very promising to explore the efficiency of gradient-based NAS in federated global communication and local updates for the RSSC.

However, there is still no relevant research exploring how to implement NAS into the federated RSSC (FedSC) to efficiently provide a personalized customized architecture for each participant. In this article, we first formulate a generalized federated RSSC framework, called FedSC. FedSC innovatively integrates the FL into RSSC, which is compatible with any manually designed architecture, network pruning strategy, and NAS. Based on the FedSC, we design a customized neural architecture search (CNAS) approach for the personalized federated problem. Specifically, each participant carries out the response to receive the supernet from the central server, and executes CANS according to the local computational capability to achieve multiparty personalized collaborative learning under the protection of the private RS data. In the proposed CNAS, the customized layer alignment regularization term is integrated for each participant to alleviate the problem of architectural nonconvergence in complex RS scenarios, and compromise to obtain the cooperative optimal solution of shallow layer globalization and deep layer personalization in the local model. Subsequently, participants upload the network weights and architectural parameters to the central server for aggregation. Compared to FL algorithms with fixed models, the proposed FedSC is able to perform distributed collaborative training globally to obtain a generalized global supernet on the one hand, and provide customized model architectures on the other hand, taking into account the arithmetic requirements of local participants and the characteristics of RS data.

Overall, the contribution of this article can be summarized in the following three aspects.

1) We formulate a novel federated distributed collaborative RSSC framework, called FedSC. FedSC enables multiparty co-training while protecting participant data

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privacy, and can be equipped with a variety of manually designed networks, network pruning strategies, and NAS methods.

2) We design the CNAS on the basis of the FedSC framework, abbreviated as $\text{FedSC}_{\text{CNAS}}$, not only to obtain a global model with comprehensive generalizability from multiparty collaborative distributed training, but also to provide each participant with a customized model that is more compatible with local RS data.
3) We compare the proposed $\text{FedSC}_{\text{CNAS}}$ with manually designed networks, network pruning strategies, and other NAS methods on three benchmark RSSC datasets. The comparative experiments and parameter analysis experiments verify the effectiveness of the proposed $\text{FedSC}_{\text{CNAS}}$.

The remainder of this article is constructed as follows. Section II briefly reviews the related work on FL and NAS for RSSC. Our proposed framework and the specific customized NAS strategy are explicitly described in Section III. Section IV reports and analyzes the experimental results. Conclusions are drawn in Section V.

II. Related Work

A. Federated Learning

FL is an emerging machine-learning approach that allows multiple devices or servers to collaboratively train a shared model while keeping the data localized, without the need to centralize the data in a single location [21], [22], [43]. The advantage of this approach over centralized machine learning is that it improves data privacy and security, as sensitive data can be kept local and does not need to be shared [44]. In general, FL primarily focuses on horizontal FL, vertical FL, and federated transfer learning [19].

Specifically, horizontal FL targets scenarios where the data features are the same or similar but the samples are different [45]. For example, clients in different regions may use the same application, but their respective datasets do not overlap. Vertical FL applies to scenarios where the samples are the same or similar, but the features are different [46]. For example, different financial institutions may have different datasets of the same customers. Federated transfer learning, on the other hand, is suitable for scenarios where neither the samples nor the features are exactly the same [47]. With the transfer learning, cross-domain knowledge can be utilized to improve learning efficiency while maintaining data privacy.

Currently, emerging trends in FL researches has focused on the exploration of NAS for personalization strategies [48], [49], [50], [51], [52]. Specifically, Liu et al. [48] designed a FL NAS algorithm based on multiobjective evolutionary optimization for medical image classification of lung and colon histopathology images. In [49], a real-time co-evolutionary NAS method called RT-FedEvoNAS is devised to optimize the model performance and reduce the local load with a dual-sampling strategy, which effectively reduces the computational and communication costs required for evolutionary optimization. Wang et al. [50] proposed an efficient graph neural network (GNN) structure search algorithm called FL-AGNNS, which seeks the best GNN architecture for local data with the federated evolutionary optimization strategy. Liu et al. [51] searched and trained optimal model architectures in parallel by employing hierarchical NAS, which divides clients into clusters based on data distribution, and mining sub-networks from pretrained hypernetworks and assigning them to specific client clusters. Similarly, Yan et al. [52] also employed the hierarchical NAS, but designing the base unit to be shared by all clients to capture universal knowledge and the personalized unit to be customized to local data.

In the field of RS, the security of RS data has received much attention recently. Xu et al. [18] discuss the security of artificial intelligence (AI) in the field of Earth science and RS, in particular the challenges faced by the widespread use of deep-learning algorithms in RS data analysis. The article systematically reviews current developments in five important aspects: adversarial attack, backdoor attack, FL, uncertainty, and explainability. Zhang et al. [24] proposed a federated deep-learning scheme to extract objects in very high-resolution (VHR) RS images with prototype matching. This scheme utilizes RS data distributed across multiple clients for collaborative learning to compensate for the scarcity of labels in local datasets. In [25], a federated target identification method is proposed to effectively improve the performance of radar target detection by means of distributed learning, self-re-weighting loss function, target reconstruction technique, and virtual drift calibration technique for the problem of data security of multiple radar sensors. Li et al. [20] proposed a FedFusion framework, which utilizes manifold learning and FL in conjunction with a dynamic feature approximation method of singular value decomposition, to achieve on-orbit fusion of multisatellite and multimodal data without the need to transmit the data back to the ground. The FedFusion co-estimates the salient manifold structure of shallow features at each client by randomly sampling local data at each client, and explicitly compresses the feature matrix into low-rank subspaces via concatenation and addition methods, which serve as feature inputs to subsequent classifiers. In [22], a generalized multiparty personalized collaborative learning framework is proposed to achieve personalized decoupling of global and local models for different RS tasks. Wang et al. [26] proposed a personalized multiparty framework for few-shot scene classification. Specifically, it incorporates a personalized adaptive distillation scheme to initialize the local model and prevent overfitting with adaptive aggregation, and proposes a multi-scale feature matching network with an internal self-attention mechanism to improve the discrimination of different RS scenarios. Besides, reinforcement learning [53], membership-privacy-preserving [54], and multisatellite networks [55] are also explored in RS image classification.

B. NAS for RS Scene Classification

NAS is an automated method for designing deep-learning model architectures, and its application in RSSC is an emerging research area [35], [37], [38]. NAS improves the performance of RSSC by exploring different network architectures in the search space to find the optimal model architecture

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[56]. Current work on NAS focuses on the search space and search strategies for RSSC [40]. The search space is the set of all possible network architectures, including different convolutional layers, pooling layers, attention mechanisms, and other network components. The search strategy determines the navigation in the search space to find the optimal architecture, and common search strategies include evolutionary algorithms, reinforcement learning, and gradient descent methods [57].

Peng et al. [36] proposed an efficient CNN architecture search framework called GPAS, which automatically discovers the optimal network structure suitable for RSSC by gradient optimization method, and constructed a large-scale cross-domain RS image dataset, VHRRS, to enhance the model's generalization ability. In [41], a framework called RS-DARTS is proposed to optimize the search process by introducing strategies such as cooperative mechanism, binarization of structural parameters, adding noise, and sampling feature channels. Ma et al. [34] proposed a novel NAS framework based on a multiobjective evolutionary algorithm called SceneNet, which is able to adaptively provide diverse architectures by balancing computational complexity and performance errors by encoding network connectivity patterns. In [58], an automated CNN architecture search method based on symbolic linear generative coding, combining stochastic search, and early stopping strategies, is proposed for RS image scene classification task. Wan et al. [39] proposed an efficient automatic search framework based on the multiobjective evolutionary algorithm, called E2SCNet, for building deep-learning network architectures suitable for RS image scene classification. E2SCNet effectively balances the accuracy and the number of parameters by constructing a diverse search space with lightweight operators and introducing a large model retention mechanism and a supernetwork weight-sharing strategy.

III. Methodology

A. Problem Statement

In FL, multiple participants (e.g., different devices, organizations, or individuals) collaboratively train a model while protecting data privacy. Each participant can store and process its private data locally without centralizing it on a central server [43]. Suppose a total of K participants are involved in the collaborative training, the optimization problem can be expressed as follows:

$$\min_{\theta}\left\{ f(\theta)=\frac{1}{K}\sum_{k=1}^{K} f_k(\theta)\right\} \tag{1}$$

where θ is the global model to be pursued.

However, in RS image processing, different participants actually prefer to obtain personalized local models that conform to local RS data. Therefore, in the proposed FedSC framework, the aim is to automatically search for the neural network architectures with the most appropriate local privacy RS dataset for each participant under a fixed search space. Based on this preliminary intent, the dataset is divided into three nonoverlapping parts: a training dataset $\mathcal{D}_{\text{Train}}$ for training the network weight ω, a validation dataset $\mathcal{D}_{\text{Val}}$ for searching the architecture parameter α, and a testing dataset $\mathcal{D}_{\text{Test}}$ for evaluation. For each participant, the privacy RS data can be represented as $\mathcal{D}_k := \{(x_i, y_i)\}_{i=1}^{N_k}$, where N_k is the size of the dataset for the kth participant. Consequently, the optimization problem for the gradient-based federated NAS can be statement as follows:

$$\min_{(\omega,\alpha)}\left\{ f(\omega,\alpha)=\frac{1}{K}\sum_{k=1}^{K} f_k(\omega,\alpha)\right\} \tag{2}$$

where $f_k(\omega,\alpha)=\sum_{i=1}^{N_k}\mathcal{L}_k(x_i, y_i; \omega_k, \alpha_k)$, ω_k and α_k are the local network weight and architecture parameter for the kth participant, respectively.

It is worth noting that the network weight and architectural parameter are optimized separately in the training and validation sets. Here, the training loss and validation loss are denoted by $\mathcal{L}_{\text{Train}}$ and $\mathcal{L}_{\text{Val}}$, respectively. Both of these two losses depend not only on the architecture parameter α but also on the network weight ω. The objective of NAS is to search the α^* that minimizes the validation loss $\mathcal{L}_{\text{Val}}(\omega^*, \alpha^*)$, where ω^* is obtained by minimizing the training loss $\omega^* = \arg\min_{\omega}\mathcal{L}_{\text{Train}}(\omega, \alpha^*)$. Actually, this is a two-level optimization problem, where α is the upper level variable and ω is the lower level variable

$$\begin{aligned}&\min_{\alpha}\mathcal{L}_{\text{Val}}\left(\omega^*(\alpha), \alpha\right),\\ &\text{s.t. } \omega^*(\alpha)=\arg\min_{\omega}\mathcal{L}_{\text{Train}}(\omega,\alpha).\end{aligned} \tag{3}$$

Overall, the (2) can be written as

$$\begin{aligned}&\min_{(\omega,\alpha)}\left\{ f(\omega,\alpha)=\sum_{k=1}^{K}\frac{1}{K} f_k\left(\omega^*(\alpha),\alpha\right)\right\}\\ &\text{s.t. } \omega^*(\alpha)=\arg\min_{\omega}\mathcal{L}_{\text{Train}}(\omega,\alpha).\end{aligned} \tag{4}$$

Based on the above problem statement, the proposed FedSC should be satisfied with the following properties.

1) *Distributed Training*: FedSC requires distributed training of models on multiple participating nodes, where each node uses local data for training and only shares model weights and architectural parameters, to protect user data privacy.
2) *Personalized Local Network*: FedSC needs to automate the search for the optimal neural architecture for each participant that is best suited to the local data under different computational resource conditions.
3) *Lightweight Model*: FedSC should be able to strike a balance between multiple objectives such as accuracy, model parameters, computational complexity, and communication overhead, to minimize search time and resource consumption searching for the lightweight and efficient models.

B. Federated NAS Framework

The process of the proposed federated NAS framework for RSSC is shown in Fig. 2. It is worth noting that unlike previous centralized learning paradigm, the private RS data of the participants are not accessible to each other in the proposed framework. First, a corresponding supernet is distributed at the central server based on the user response, where the search

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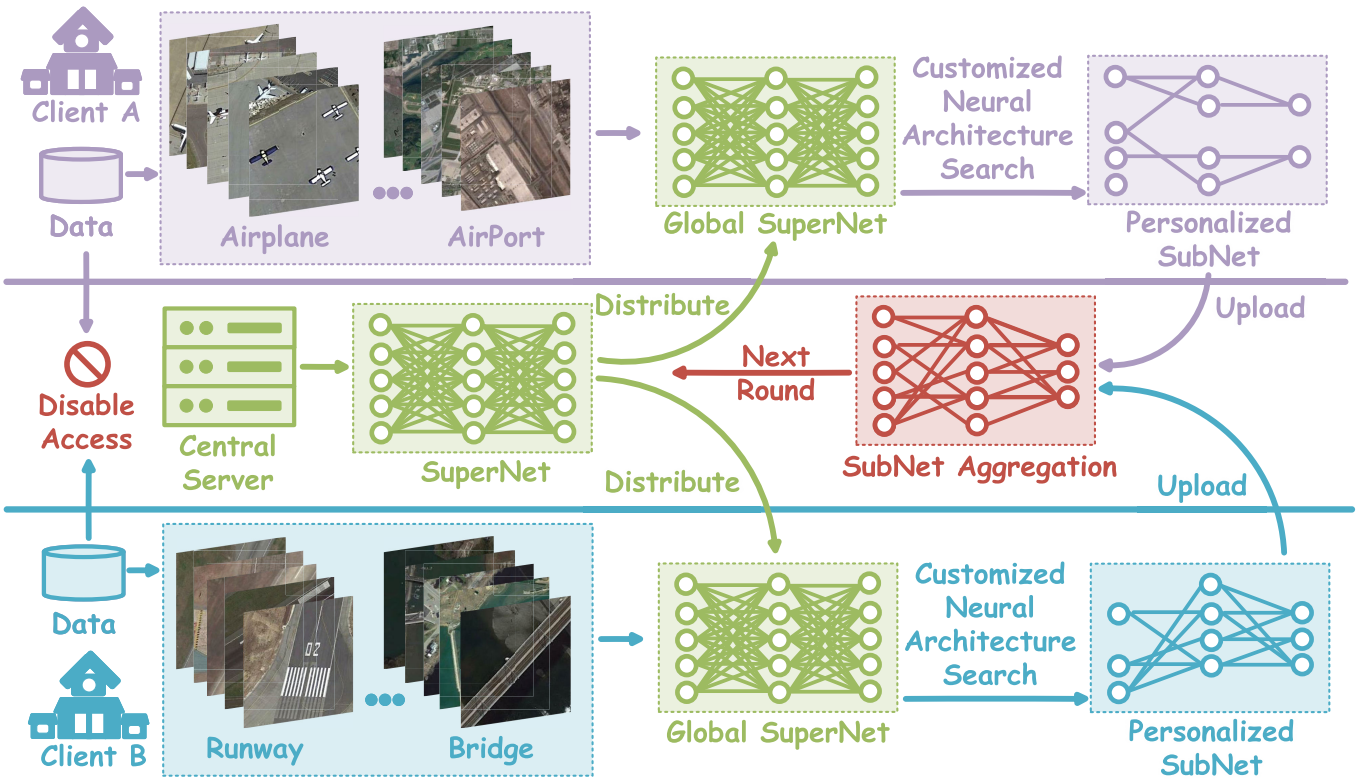


Fig. 2. Schematic of the proposed federated NAS framework. We show the flow of one global communication round. First, the central server distributes a supernet to different participants based on their responses, and each participant performs the personalized NAS locally to obtain the network weight and architectural parameter based on the private RS data, which are subsequently uploaded to the central server for aggregation.

space adopted the gradient-based NAS paradigm [32], [33] and consists of normal cell and reduction cell.

1) *Normal Cell*: Normal cell is the standard unit in the search space for transforming and combining features while maintaining constant feature size. It usually contains multiple nodes, each of which is a combination of preorder nodes with various operations (e.g., depth-wise separable convolution, dilated convolution, direct connection, skip connection, max pooling, and average pooling).
2) *Reduction Cell*: Reduction cell is specialized to reduce the size of the feature map, and is typically inserted after every few normal cells in the search space to achieve downsampling. Operations in reduction cell may result in halving the width or height of the feature map compared to normal cell. Reduction cell contributes to a reduction in computation and can increase the receptive field, facilitating the capture of global features.

Each cell can be interpreted as a directed acyclic graph and each node is connected to the preceding node, where nodes can be considered as feature maps, and edges represent the operations employed. Specifically, in each cell, $\mathcal{N}^{(i)}$ denotes the potential feature representation corresponding to the ith node, and $o^{(i,j)}$ represents the operation adopted from the ith node to the jth node, and $\mathcal{O}$ is all the candidate operations. Each node is connected to all previous nodes and the result is summed to obtain the feature graph of the jth node. In the search space, each edge selects an operation to relax the discrete search space. Therefore, each node and edge is represented as follows:

$$\mathcal{N}^{(j)} = \sum_{i<j} o^{(i,j)}\left(\mathcal{N}^{(i)}\right) \tag{5}$$

$$\bar{o}^{(i,j)}(\mathcal{N}) = \sum_{o\in\mathcal{O}} \frac{\exp\left(\alpha_o^{(i,j)}\right)}{\sum_{o'\in\mathcal{O}} \exp\left(\alpha_{o'}^{(i,j)}\right)} o(\mathcal{N}) \tag{6}$$

where $\alpha^{(i,j)}$ denotes the weight of the operation between the ith and the jth feature map, which is the network structure parameter to be searched afterward. After the training is completed, the edge with the highest probability is obtained from all the edges, denoted as

$$o^{(i,j)} = \arg\max_{o\in\mathcal{O}} \alpha_o^{(i,j)}. \tag{7}$$

Assume that there are T global communication rounds, and the proposed FedSC$_{\text{NAS}}$ process is shown in Algorithm 1. After the participants receive the supernet, personalized NAS is implemented based on local arithmetic combined with their privacy RS data. In the tth global communication round for the kth participant, the respective network weight ω_k^t and architectural parameter α_k^t are updated as

$$\begin{aligned} &\min_{\alpha_k^t} \mathcal{L}_{\mathcal{D}_k^{\text{Val}}}\left(\omega_k^{t*}\left(\alpha_k^t\right), \alpha_k^t\right) \\ &\text{s.t. } \left(\omega_k^{t*}, \alpha_k^t\right) = \arg\min_{\omega_k^t} \mathcal{L}_{\mathcal{D}_k}^{\text{Train}}\left(\omega_k^t, \alpha_k^t\right). \end{aligned} \tag{8}$$

Subsequently, each participant uploads the novel network weight and architectural parameter to the central server for

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Algorithm 1 FedSC$_{\text{NAS}}$ Framework

Input: K: the number of participants, T: the number of global communication rounds, E: the number of local update rounds, $\mathcal{D}_k$: the local RS data of kth participant.

Output: α^* the best architecture parameter, ω^*: the best network weight.

for $t = 1$ to T **do**
  Send the global architecture parameter ω^t and global network weight α^t to K participants.
  for $k = 1$ to K **do**
    Update the local architecture parameter α_k^t and the local network weight ω_k^t with $\mathcal{D}_k$ based on Equation (12).
    Upload the updated α_k^t and ω_k^t to the central server.
  end for
  Aggregate the network weight ω_{t+1} and architectural parameter α_{t+1} based on Equations (9)–(10).
  $t \leftarrow t + 1$.
end for

aggregation as shown below

$$\omega_{t+1} = \frac{1}{K}\sum_{k=1}^{K}\omega_k^t, \tag{9}$$

$$\alpha_{t+1} = \frac{1}{K}\sum_{k=1}^{K}\alpha_k^t \tag{10}$$

and this process continues until the number of global communication rounds reaches the maximum number of rounds set by the participants.

C. Personalized Customized NAS Strategy

The current gradient-based NAS [32], [33] in the FedSC framework suffers from unstable performance collapse, especially the architecture obtained from the search does not converge in the presence of different complicated RS scenes. Recently, several related researches [35], [36], [39] have been targeted to improve on the search strategy, but only for the centralized machine-learning paradigm, and have not been designed specifically on the FL framework. In this article, we proposed a novel customized NAS (CNAS) strategy for FedSC to provide a stable personalized architecture for each participant.

Specifically, the gradients of the different layers can be almost orthogonal due to the weight sharing framework, which implies that they have virtually synergy in architecture selection, resulting in some layers especially those close to the output layer suffering from unfair practices in selecting operations. Whereas in FL paradigm, the shallow layers of the network are global because the network aggregation for different architectures enables the global model to capture more generalizable features, while the deep layers of the network are specific to local RS data characteristics.

In the gradient-based NAS, the optimal cellular structure of the searched network located in the shallow layers usually consists of convolutional operations, while the deeper layers are composed of skip connections. This is because skip-connection facilitates the gradient propagation to deeper cells of the architecture, while convolution provides more complex features to the shallow layers. In addition, due to gradient vanishing, the average value of the gradient for skip-connection operations is large, and the gradient value decreases as the depth of the network increases. As a result, the negative value of the objective function tends to be smaller for skip-connection operations, further exacerbating the performance collapse problem.

Therefore, we employed the customized layer alignment strategy for different participants in CNAS. To measure the correlation between the gradients of the layers and the architectural parameters in each participant, the layer alignment metric $\Omega(\omega_k, \alpha_k)$ for kth participant is implemented by averaging the normalized dot products of all pairs of layers as follows:

$$\Omega(\omega_k, \alpha_k) = \frac{1}{\binom{L_k}{2}}\sum_{\ell_k<\ell_k'}\frac{\nabla_{\ell_k^\beta}\mathcal{L}(\omega_k,\alpha_k)^T\nabla_{\ell_k'^\beta}\mathcal{L}(\omega_k,\alpha_k)}{\|\nabla_{\ell_k^\beta}\mathcal{L}(\omega_k,\alpha_k)\|_2\|\nabla_{\ell_k'^\beta}\mathcal{L}(\omega_k,\alpha_k)\|_2} \tag{11}$$

where L_k is the number of layers in the network, $\nabla_{\ell_k^\beta}$ denotes the gradient of the corresponding architectural parameter β at ℓ_kth layer, $\|\cdot\|_2$ represents the Euclidean norm, and $\sum_{\ell_k<\ell_k'}$ means that the summation is performed over all possible pairs of layers, where ℓ_k and ℓ_k' denote different layers.

Due to the weight-sharing structure, the $\Omega(\omega_k, \alpha_k)$ has a value close to zero, resulting in the search yielding architectures that converge only at the saturation point of the softmax function. Because of the insufficient alignment of the deeper network, the optimal architectures for each layer differ significantly, with deep layers tending to choose skip connections more frequently, which gives the skip connection operation a distinct advantage during the search. As the search iterates, the architectural parameters corresponding to the skip-connection operation will be more than the architectural parameters of the convolution operation, which ultimately leads to an increase in the proportion of skip-connect operations chosen at the saturation point, further exacerbating the performance breakdown problem. In other words, $\Omega(\omega_k, \alpha_k)$ is defined as the regularization term that needs to be maximized, but when the dimension of α_k is large, it will be concentrated on operations with relatively large gradients, which means that although the angle between the layer gradients decreases, only a small portion of the operations will have the same sign of the gradient, which again leads to degradation of performance.

Based on the above definition, the first step of the proposed FedSC$_{\text{CNAS}}$ framework is for each participant to receive the supernet distributed by the central server based on the response. Subsequently, a customized NAS is performed based on the local RS data, and the gradient $\nabla_{\ell_k^\beta}\mathcal{L}(\omega_k, \alpha_k)$ of the corresponding architectural parameter α is obtained by performing gradient computation for each layer, which is done by normalizing the architectural parameters with the softmax function $\sigma(\alpha_k)$, and the gradient of the loss function $\mathcal{L}_k$ with respect to these normalized parameters is computed. For each pair of layers ℓ_k and ℓ_k', the dot product of their gradients is computed, measuring the similarity of the two gradient

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TABLE I
DETAILED INFORMATION OF FOUR FEW-SHOT RSSC DATASETS

Dataset	Year	Publisher	Categories	Images/Category	Total Images	Size	Resolution (m)
AID [59]	2016	Huazhong University of Science and Technology & Wuhan University	30	220∼420	10,000	600 × 600	0.5∼8
NWPU-RESISC45 [1]	2017	Northwestern Polytechnical University	45	700	31,500	256×256	0.2∼30
PatternNet [60]	2018	Wuhan University	38	800	30,400	256×256	0.062∼4.693

TABLE II
COMPARISON RESULTS BETWEEN THE "BASELINE" AND THE PROPOSED CNAS

Search Space	AID		NWPU-RESISC45		PatternNet	
	baseline	**CNAS (Ours)**	baseline	**CNAS (Ours)**	baseline	**CNAS (Ours)**
S1	67.09	**68.31** (↑ **1.22**)	65.16	**66.22** (↑ **1.06**)	92.44	**94.19** (↑ **1.75**)
S2	64.32	**65.06** (↑ **0.74**)	58.76	**62.61** (↑ **3.85**)	91.49	**93.32** (↑ **1.83**)
S3	62.22	**62.72** (↑ **0.50**)	59.69	**66.90** (↑ **7.21**)	91.26	**92.35** (↑ **1.09**)
S4	69.43	**69.81** (↑ **0.38**)	68.29	**71.87** (↑ **3.58**)	88.56	**94.52** (↑ **5.96**)

TABLE III
COMPARISON RESULTS OF AA IN % AND PARAMETERS (PARAMS.) IN M OF THE GLOBAL MODEL, WHERE THE "†" DENOTES THE NETWORK PRUNING STRATEGY BASED ON THE VGG-16 [63] ARCHITECTURE

Method	AID				NWPU-RESISC45				PatternNet			
	IID		Non-IID		IID		Non-IID		IID		Non-IID	
	AA	Params.	AA	Params.	AA	Params.	AA	Params.	AA	Params.	AA	Params.
Manually Designed Network												
FedSC$_{\text{AlexNet}}$ [64]	53.73	57.19	51.23	57.19	54.70	57.19	50.54	57.19	87.62	57.19	84.96	57.19
FedSC$_{\text{ResNet18}}$ [65]	57.12	11.19	56.63	11.19	55.02	11.19	55.00	11.19	92.73	11.19	89.84	11.19
FedSC$_{\text{ResNet50}}$ [65]	57.72	23.57	54.98	23.57	57.70	23.57	56.67	23.57	91.80	23.57	88.33	23.57
FedSC$_{\text{MobileNetV2}}$ [66]	58.46	2.26	53.79	2.26	62.37	2.26	56.10	2.26	91.92	2.26	89.37	2.26
Network Pruning Strategy												
FedSC$_{\text{ThiNet}^\dagger}$ [27]	69.18	4.58	61.50	4.04	69.67	4.58	60.39	4.05	93.45	4.13	88.05	3.85
FedSC$_{\text{Taylor Pruning}^\dagger}$ [28]	69.05	3.24	61.15	2.99	70.55	3.24	60.28	3.00	93.11	3.01	87.53	2.84
FedSC$_{\text{Network Slimming}^\dagger}$ [29]	68.63	3.85	61.22	3.52	70.07	3.86	60.35	3.53	92.98	3.61	87.63	3.35
FedSC$_{\text{SEP}^\dagger}$ [30]	68.85	3.57	61.43	3.23	69.92	3.58	60.54	3.24	93.26	3.35	87.85	3.07
FedSC$_{\text{L1 Pruning}^\dagger}$ [31]	69.24	3.37	61.15	3.09	70.88	3.38	60.32	3.10	93.15	3.20	87.59	2.95
Neural Architecture Search												
FedSC$_{\text{Darts}}$ [32]	72.83	0.17	68.43	0.15	74.31	**0.11**	65.29	0.13	95.53	0.16	88.56	**0.11**
FedSC$_{\text{DartsV2}}$ [32]	72.67	**0.12**	68.43	0.16	75.02	**0.11**	65.91	0.13	95.34	**0.11**	92.91	0.12
FedSC$_{\text{MiLeNas}}$ [33]	73.60	**0.12**	68.73	**0.14**	77.22	0.12	62.52	**0.11**	94.18	**0.11**	92.49	0.14
FedSC$_{\textbf{CNAS}}$	**75.46**	0.15	**69.81**	0.16	**81.33**	0.12	**71.87**	0.15	**96.07**	0.12	**94.52**	0.12

vectors. Subsequently, the dot product is normalized with the mode of the gradient of each layer, ensuring that the metric is not affected by the size of the gradient. Finally, the layer alignment metric is obtained by computing the average of the normalized dot products of all pairs of layers. Therefore, the (8) is updated as follows:

$$\begin{aligned}&\min_{\alpha_k^t} \mathcal{L}_{\mathcal{D}_k^{Val}}\left(\omega_k^{t*}\left(\alpha_k^t\right), \alpha_k^t\right)\\&\text{s.t. } \left(\omega_k^{t*}, \alpha_k^t\right)\\&= \arg\min_{\omega_k^t}\left\{\mathcal{L}_{\mathcal{D}_k}^{Train}\left(\omega_k^t, \alpha_k^t\right) + \lambda_k \Omega^{Train}(\omega_k, \alpha_k)\right\} \qquad (12)\end{aligned}$$

where λ_k is the weight parameter of the layer alignment metric $\Omega(\omega_k, \alpha_k)$.

IV. EXPERIMENTAL RESULTS

A. Datasets

To validate the effectiveness of our proposed framework, three benchmark RSSC datasets are employed with specific information as summarized in Table I.

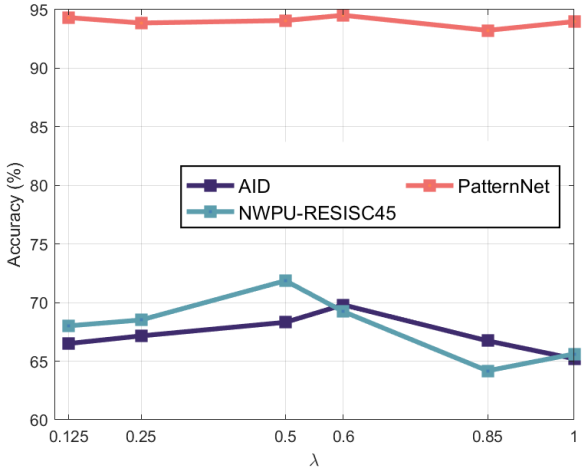


Fig. 3. Parameter analysis experiment of λ.

1) Aerial Image Dataset [59]: The aerial image dataset (AID) dataset is a large-scale dataset for RSSC, released by Huazhong University of Science and Technology and Wuhan

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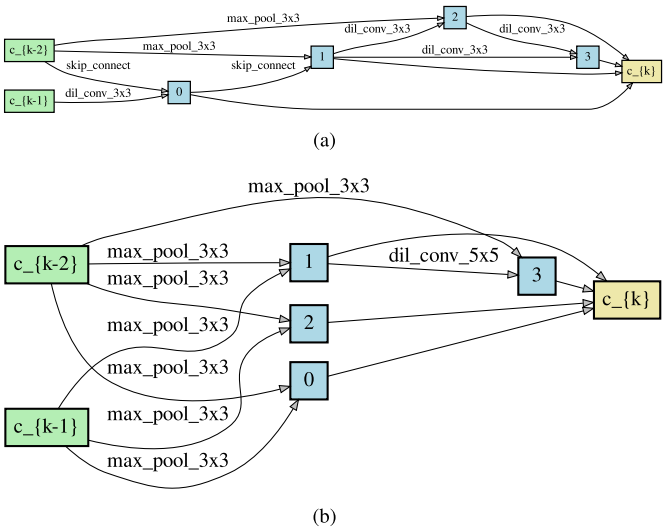


Fig. 4. Best normal cell and reduction cell of the proposed CNAS in the AID dataset. (a) Best normal cell. (b) Best reduction cell.

University in 2016. The dataset contains 30 different RS scene categories covering a wide range of RS images from natural landscapes to man-made structures, with an image size of 600 × 600 pixels and a spatial resolution of 0.5–8 m, and containing about 10 000 images, which provides researchers with sufficient data to train and test their models. The AID dataset aims to provide a standardized platform for researchers to evaluate and compare different RS image classification algorithms.

2) NWPU-RESISC45 Dataset [1]: The NWPU-RESISC45 dataset was released by the Northwestern Polytechnical University in 2017. The dataset contains 45 categories with 700 images in each category, totaling 31 500 RS scene images of 256 × 256 pixels. The spatial resolution of this dataset ranges from approximately 0.2–30. The NWPU-RESISC45 dataset has become an important benchmark dataset in the field of RS image classification due to its large scale, diversity and high quality, which is of great significance in promoting the development of RS image processing technology.

3) PatternNet Dataset [60]: The PatternNet dataset was released by Wuhan University in 2018. This dataset contains 38 categories, each containing 800 images with a size of 256 × 256 pixels and a spatial resolution of 0.062–4.693 m. The total of 30 400 images in the dataset have high spatial resolution and cover a wide range of different RS scenarios, and the diversity and large-scale nature of the dataset make it a valuable resource in the field of RS image recognition.

B. Implementation Details

In the experiments, we implemented all the experiments on a dual NVIDIA GeForce RTX 4090 with CUDA version 12.4. Consistent with most FL -related studies [23], we set the number of participants to ten and the number of global communication rounds to 100. In terms of data partitioning, we report both independent identically distributed (IID) and Non-IID with Dirichlet distribution. Keeping in line with most RSSC studies [61], [62], the input image is cropped to 84 × 84 pixels. Besides, we set the local batch size to 16 and

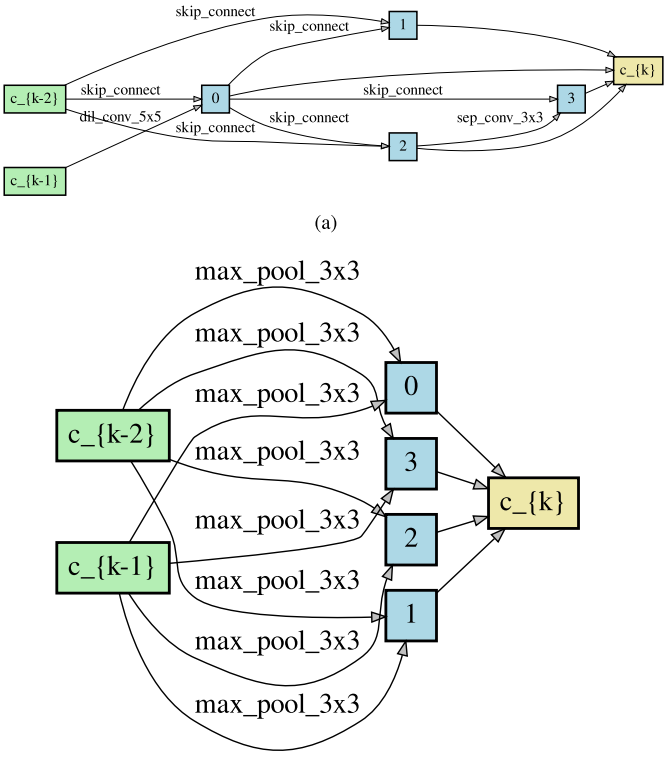


Fig. 5. Best normal cell and reduction cell of the proposed CNAS in the NWPU-RESISC45 dataset. (a) Best normal cell. (b) Best reduction cell.

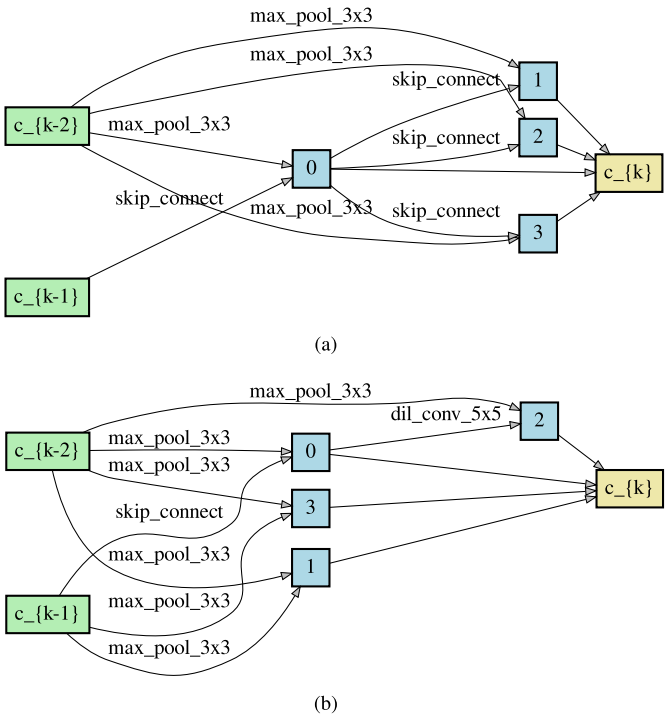
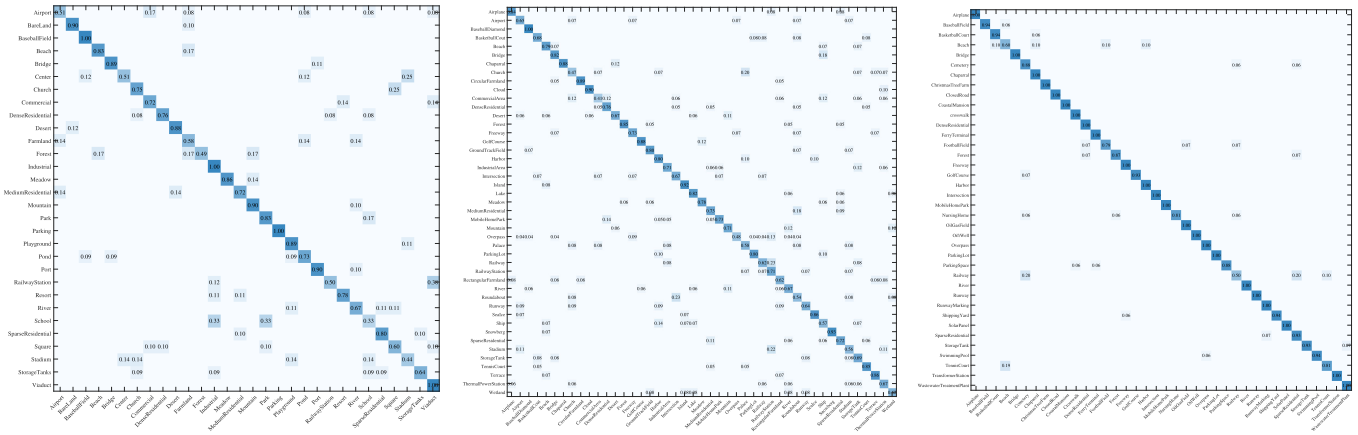


Fig. 6. Best normal cell and reduction cell of the proposed CNAS in the PatternNet dataset. (a) Best normal cell. (b) Best reduction cell.

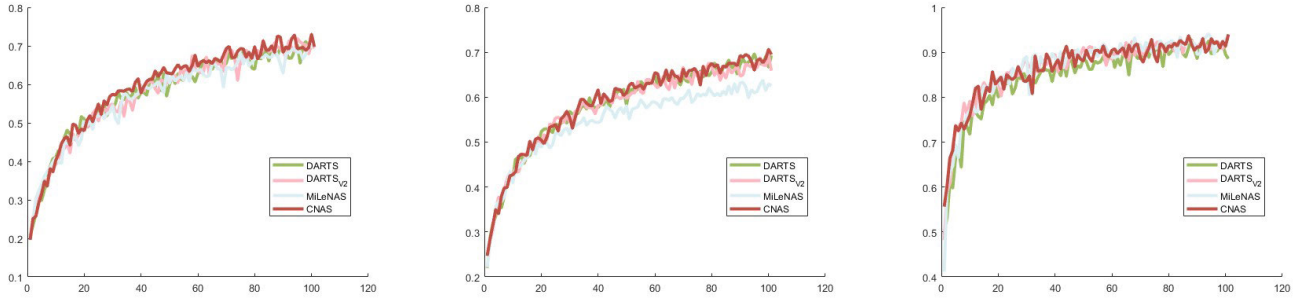
learning rate to 2.5 e^{-2}. The weight decay parameter of the ADAM is 3 e^{-4}. For the parameters of the architecture search phase, we use the dynamic learning rate by learning the scheduler and set the minimum learning rate to 1 e^{-3}. The architectural learning rate and architectural weight decay are

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(a) (b) (c)

Fig. 7. Confusion matrix of the global model. (a) AID dataset. (b) NWPU-RESISC45 dataset. (c) PatternNet dataset.



(a) (b) (c)

Fig. 8. Convergence curves for local model of one participant. (a) AID dataset. (b) NWPU-RESISC45 dataset. (c) PatternNet dataset.

set to 3 e^{-4} and 1 e^{-3}, respectively. In terms of experimental results, we report the combined metrics of both model accuracy and the parameters of model.

C. Experiment on the FedSC$_{CNAS}$

We validate the performance of the proposed CNAS against the "baseline" in different search space as shown in Table II, where the "baseline" deserves to be a simple gradient descent NAS combined to the federated RS scene categorization framework only, and the search space stays consistent with studies [32], [33]. It is worth noting that the search space S1 to S4 becomes more complex, and the corresponding selectable operation operators also increases. The experimental results show that CNAS performance is not better in more complex search spaces. In simpler search spaces, the CNAS can traverse all possible architectures more quickly, making it easier to find high-performance architectures. Whereas a complex search space offers more architectural choices, which may contain potentially high-performance architectures, but also increases the difficulty and time of the search. However, when the search space is sufficiently complete in terms of available operations, CNAS is able to compute the gradient correlation between all pairs of layers to obtain a superior architecture from a global perspective. This is because the layer alignment metric is obtained by averaging the normalized dot product over all possible pairs of layers, which reflects the average correlation of the gradients between the different layers, so its reflection becomes more significant as the search space becomes more complicated. As exemplified in the experimental results, the proposed CNAS obtain the best results in the most complex search space (i.e., S4) for all datasets. To ensure fairness in subsequent experiments, we adopted the S4 search space for all the NAS-based methods in comparison.

Besides, we compare the effect of different values of regularization weights λ on the proposed FedSC$_{\text{CNAS}}$ framework as shown in Fig. 3, which are taken as 0.125, 0.25, 0.5, 0.6, 0.85, and 1, respectively. If the value of regularization term $\Omega(\omega, \alpha)$ is large, it indicates that the gradients of the different layers are highly aligned in direction, on the contrary, it suggests that the gradients are almost orthogonal. λ is a hyperparameter that controls the degree of influence of the regularization term in the optimization process, the value of λ determines the strength of the constraints imposed by the regularization term on the optimization process. A larger value of λ means that the regularization term dominates the loss function, while a smaller value of λ indicates that the model is more focused on minimizing the original loss function. The experimental results show that too large or too small a value of λ will make the performance of CNAS degraded, so a moderate value is most appropriate.

D. Compared With Other Methods

To verify the effectiveness of the proposed CNAS, the following methods are employed for comparison, i.e., manually

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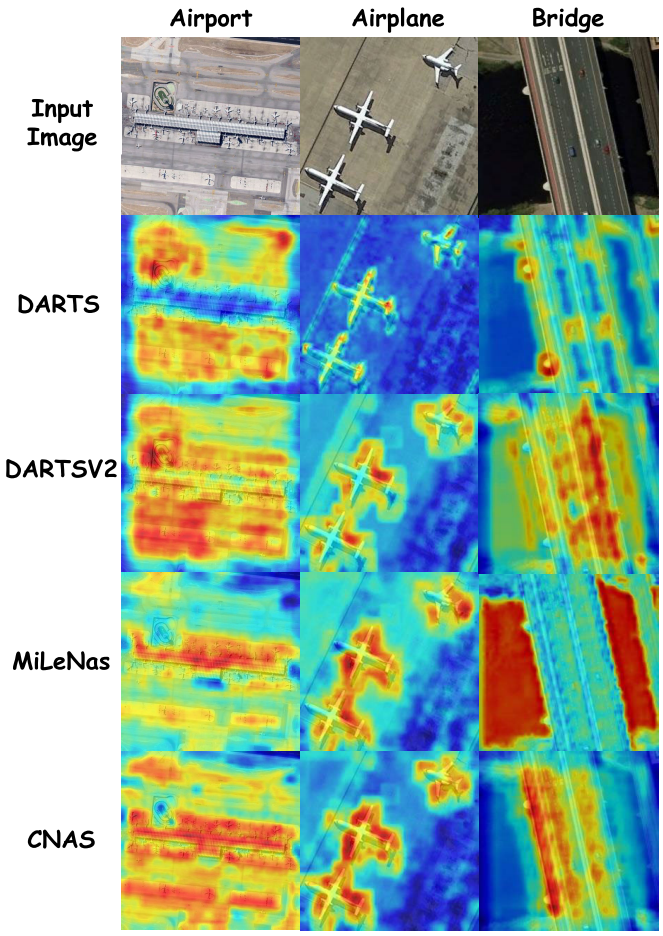


Fig. 9. Comparison of heatmap visualization of some cases for models obtained from different NAS.

designed network: AlexNet [64], ResNet18 [65], ResNet50 [65], and MobileNetV2 [66], network pruning strategy: ThiNet [27], Taylor Pruning [28], Network Slimming [29], SEP [30], and L1 Pruning [31], and NAS: Darts [32], DartsV2 [32], and MiLeNas [33]. It is worth noting that the search space of the NAS-based methods are guaranteed to be consistent for fairness, and the results of all the methods are reported from the best combination of their parameters.

Table III gives a comparison on the average accuracy (AA) and number of parameters of the global model in FedSC framework in terms of both IID and Non-IID. It can be concluded from the experimental results that compared to manually designed networks and fixed network pruning strategies that need to rely on the experience of experts, the NAS-based approaches obtain a crushing lead in the number of parameters of the model, and achieves better performance while discarding irrelevant weights. Specifically, manually designed network requires professional expertise to design the optimal architecture, and it is difficult to adapt the variations in different RS datasets, e.g., the optimal manually designed architectures on the AID, NWPU-RESISC45, and PatternNet datasets are ResNet18, AlexNet, and MobileNetV2, respectively. Although the network pruning strategy can reduce the number of model parameters and computational complexity to a certain extent, it is usually a postprocessing step based on an existing redundant model, which may not achieve globally optimal resource utilization. While the NAS approaches can obtain the optimal or suboptimal network architectures in large-scale architecture space with an automated searching process, and reduces the reliance on expert experience. In addition, the proposed CNAS can customize the optimization objectives according to the demands of specific application scenarios, e.g., low power requirements on mobile devices, or high-performance requirements in cloud computing, and search for the most suitable network architecture.

Moreover, we also provide the best normal cell and reduction cell of the proposed CANS on the three benchmark datasets as shown in Figs. 4–6. It can be observed that the proposed CNAS can combine the characteristics of different RS scenarios to search an architecture adapted to local privacy data, such as for the NWPU-RESISC45 dataset with complex scenarios, the CNAS adopts the combination of convolution operation and maximum pooling operation to utilize the feature extraction capability of convolution and the dimensionality reduction capability of pooling, and plays an important role in maintaining the flow of the gradient and reducing the disappearance of the gradient through the skip connection. The role of the skip connection is important in maintaining the gradient flow and reducing the gradient disappearance.

Fig. 7 provides the confusion matrix of the global model from the proposed CNAS. The confusion matrix is an important metric for evaluating the performance of a classification model that provides detailed classification results. In the AID dataset, which is the worst performing data in CNAS, we noticed that some relatively diverse and complex scenarios such as “airport,” “forest,” “school”, and “stadium” are not well overcome by the optimal architecture searched by CNAS. It is attributed to the fact that the weights of all candidate operations are updated through a soft parameterization in the gradient-based NAS paradigm, which can lead to a search process that is biased toward particular operations that may not necessarily be the best choice in the end.

For further analysis, Fig. 8 gives a comparison of the convergence curves of different NAS methods for the participant local model. Compared to other gradient-based NAS methods, the proposed CNAS is relatively stable in training the cooperative optimization of architectural parameters and network weights during the search process, especially on complex tasks and large-scale datasets. Besides, we also provide the comparison of heatmap visualization of some cases for models obtained from different NAS as shown in Fig. 9. It can be obtained from the case study that the model obtained from CNAS performs much better for capturing the key information about complex scenarios.

V. Conclusion

In this article, we present the problem definition, concrete implementation, and specific algorithms of the proposed $\text{FedSC}_{\text{CNAS}}$, a multiparty collaborative distributed framework specifically designed for RSSC, which achieves global FL in the protection of user-sensitive RS information, and local model customization in data heterogeneity. We validate the effectiveness and generalizability of the proposed FedSC

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framework in different hand-designed networks, network pruning strategies, and NAS approaches. Analysis and comparative experiments on three RSSC datasets, AID, NWPU-RESISC45, and PatternNet, validate that our proposed $\text{FedSC}_{\text{CNAS}}$ not only obtains globally generalizable model in the multiparty distributed training paradigm, but also provides a customized search architecture for each participant.

In the future, we intend to generalize FL and NAS for other RS tasks such as change detection, target identification, objective segmentation, and other modalities of RS data such as hyperspectral, SAR, and point cloud. Besides, the expensive time cost of NAS warrants further optimization in future work.

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